

## 2. Google Kubernetes Engine Pipeline using Cloud Build

```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-03-a83160aaeca9&pli=1&show=ide%2Cterminal

Cloud Shell Editor

Walkthrough: Setup Code OSS Web

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-03-a83160aaeca9.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_00_21b18e955805@cloudshell:~ (qwiklabs-gcp-03-a83160aaeca9)$ curl -LO https://raw.githubusercontent.com/Itsabhishek7py/GoogleCloudSkillsboost/refs/heads/main/Google%20Kubernetes%20Engine%20Pipeline%20using%20Cloud%20Build/abhishek.sh
sudo chmod +x abhishek.sh
./abhishek.sh
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  5759  100  5759    0     0  15966      0 --:--:-- --:--:-- --:--:-- 15997

Welcome to Dr Abhishek's Cloud Tutorials! ☁️
Subscribe to the channel: https://www.youtube.com/@drabhishek.5460/videos

Lo Done! [[]]
Your active configuration is: [cloudshell-16715]
Updated property [compute/region].
Operation "operations/acat.p2-880309350907-648fe7f3-e900-4e56-969d-333b958e22e2" finished successfully.
Create request issued for: [my-repository]
Waiting for operation [projects/qwiklabs-gcp-03-a83160aaeca9/locations/us-west1/operations/745a8d57-8e38-45ba-bc41-d363fdb486f0] to complete...done.
Created repository [my-repository].
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster hello-cloudbuild in us-west1... Cluster is being configured...working
```

```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-03-a83160aaeca9&pli=1&show=ide%2Cterminal

Cloud Shell Editor

Walkthrough: Setup Code OSS Web x

Extracting ~/Downloads/webi/gh/2.81.0/gh_2.81.0_linux_amd64.tar.gz
Installing to ~/.local/opt/gh-v2.81.0/bin/gh

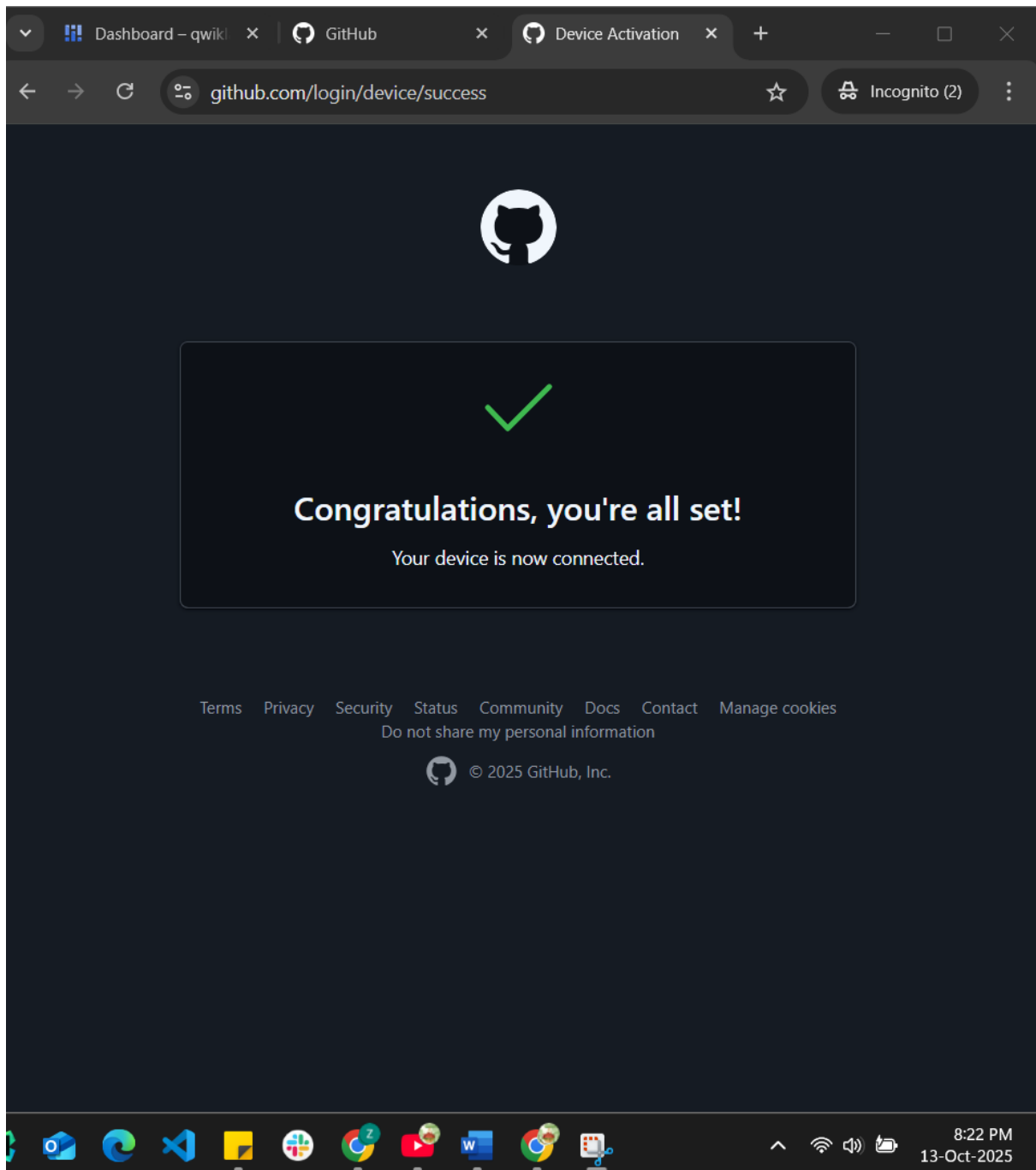
Installed 'gh v2.81.0' as ~/.local/bin/gh

Edit ~/.config/envman/PATH.env to add:
~/.local/bin

>>> ACTION REQUIRED <<<
Copy, paste & run the following command:
source ~/.config/envman/PATH.env
(newly opened terminal windows will update automatically)

? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: 8F5B-658E
Press Enter to open https://github.com/login/device in your browser...
https://github.com/login/device
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
! Authentication credentials saved in plain text
✓ Logged in as BSSE23058
BSSE23058
BSSE23058
student-00-21b18e955805@qwiklabs.net
GraphQL: Name already exists on this account (createRepository)
GraphQL: Name already exists on this account (createRepository)
Copying gs://splsgsp1077/gke-gitops-tutorial-cloudbuild/CONTRIBUTING.md to file://hello-cloud
build-app/CONTRIBUTING.md
```



```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-03-a83160aaeca9&pli=1&show=ide%2Cterminal

Cloud Shell Editor

Walkthrough: Setup Code OSS Web X

(qwiklabs-gcp-03-a83160aaeca9) X Gemini CLI X + X

hint: See the 'Note about fast-forwards' in 'git push --help' for details.
# github.com:22 SSH-2.0-1a4a900
[master d39abal] Adding known_host file.
1 file changed, 1 insertion(+)
create mode 100755 known_hosts.github
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 765 bytes | 765.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/BSSE23058/hello-cloudbuild-app
cceb290..d39abal master -> master
% Total      % Received % Xferd  Average Speed   Time    Time     Time  Current
   0      0     0     0     0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 3352 100 3352     0     0 47676     0  --:--:-- --:--:-- --:--:-- 47676
[master 1db9b26] Trigger CD pipeline
1 file changed, 76 insertions(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.60 KiB | 546.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/BSSE23058/hello-cloudbuild-app
d39abal..1db9b26 master -> master
✅ Lab is now Completed!
👤 Thanks for using Dr Abhishek's Cloud Tutorials!
👉 Subscribe here: https://www.youtube.com/@drabhishek.5460/videos
student_00_21b18e955805@cloudshell:~ (qwiklabs-gcp-03-a83160aaeca9) $
```

Create trigger – Cl xGitHub xDevice Activation x+ x

console.cloud.google.com/cloud-build/triggers;region=u... Incognito (2)

Google Cloud

Cloud Build / Triggers

Create trigger

Repository service

☒ Cloud Build repository service

☐ Developer Connect

Repository generation

☒ 1st gen

☐ 2nd gen

Repository \*

Filter Type to filter

No repositories in region

☐ Invert Regex

Show included and ignored

Configuration

Type

☒ Cloud Build configuration

Connect repository

Region: us-west1

☒ S

Filter Type to filter

1 of 4 selected

☐ BSSE23058/DBMS-Group-5

☐ BSSE23058/Group-B\_SE\_Project

☒ BSSE23058/hello-cloudbuild-app

☐ BSSE23058/hello-cloudbuild-env

Edit repositories on GitHub

Cancel OK

☐ I understand that GitHub content for the selected repositories will be transferred to this GCP project to provide the connected service. Principals with access to this GCP project with sufficient permissions will be able to create and run triggers on these repositories, based on transferred GitHub content. I also understand that content from all GitHub App triggers in this GCP project may be transferred to GitHub in order to provide functionality like showing trigger names in GitHub build results. This will apply to all existing and future GitHub App triggers in this project. [Learn more](#)

Connect

8:27 PM

13-Oct-2025

Create trigger – Cl...xGitHubxDevice Activationx+

console.cloud.google.com/cloud-build/triggers;region=u...Incognito (2)

Google Cloudqwiklabs-gcp-03-a83160aaeca9

Cloud Build / Triggers / Create trigger

Create trigger

Repository service

☒ Cloud Build repositories

☐ Developer Connect **Preview**  
Developer Connect is a new service that provides a streamlined way for you to manage connections to source code management platforms outside of Google Cloud

Repository generation

☒ 1st gen

☐ 2nd gen

Repository \*  
BSSE23058/hello-cloudbuild-app (GitHub App)

Select the repository to watch for events and clone when the trigger is invoked

Branch \*  
.\*

Trigger only for a branch that matches the given regular expression [Learn more](#)

☐ Invert Regex

Matches the branch: master

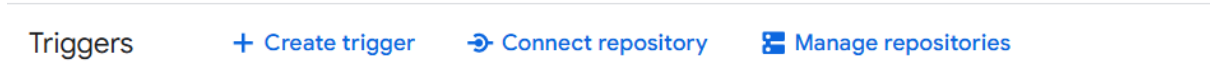
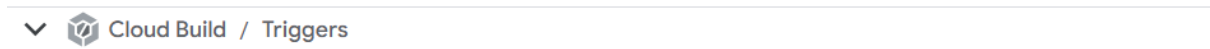
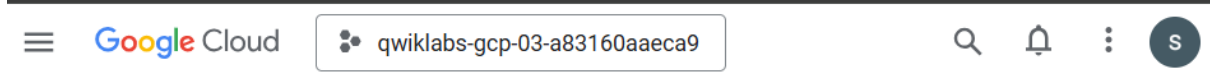
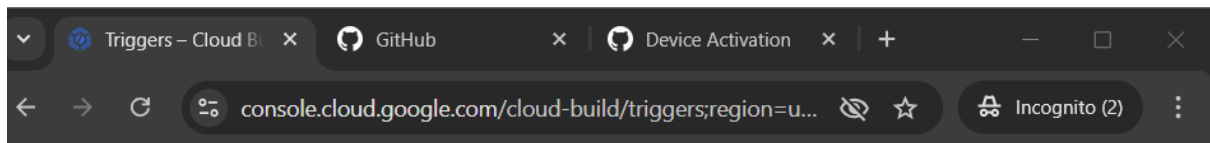
[Show included and ignored files filters](#)

Configuration

Type

Connected repository  
"BSSE23058/hello-cloudbuild-app"

8:27 PM  
13-Oct-2025



A Cloud Build trigger automatically starts a build whenever you make any changes to your source code or some other incoming event.

| Filter Enter property name or value |          |             |                                                |       |     | ? |   |
|-------------------------------------|----------|-------------|------------------------------------------------|-------|-----|---|---|
| Name ↑                              | Region   | Description | Repository                                     | Event | Run |   |   |
| <a href="#">hello-cloudbuild</a>    | us-west1 | —           | <a href="#">BSSE23058/hello-cloudbuild-app</a> | Push  | Run |   | ⋮ |

'hello-cloudbuild' has been created on  
BSSE23058/hello-cloudbuild-app



Create trigger – Cl xGitHub xDevice Activation x+ x

console.cloud.google.com/cloud-build/triggers/add;regi... Incognito (2)

Google Cloud

Cloud Build / Triggers

Create trigger

Repository generation

- 1st gen
- 2nd gen

Repository \*

- Filter Type to filter
- BSSE23058/hello-cloud
- Invert Regex
- Show included and ignored

Configuration

- Type
  - Cloud Build configuration
  - Dockerfile
  - Buildpacks
- Location
  - Repository

Connect repository

Region: us-west1

1 Select source code management provider

2 Authenticate

3 Select repository

Select the GitHub repositories to connect to Cloud Build. Principals with access to this Google Cloud project will be able to create and run triggers on these repositories.

GitHub Account \*

BSSE23058

Repository \*

BSSE23058/hello-cloudbuild-env

☒

 I understand that GitHub content for the selected repositories will be transferred to this GCP project to provide the connected service. Principals with access to this GCP project with sufficient permissions will be able to create and run triggers on these repositories, based on transferred GitHub content. I also understand that content from all GitHub App triggers in this GCP project may be transferred to GitHub in order to provide functionality like showing trigger names in GitHub build results. This will apply to all existing and future GitHub App triggers in this project. [Learn more](#)

Connect



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Triggers – Cloud B

GitHub

Device Activation

+

←

→

↺

console.cloud.google.com/cloud-build/triggers;region=u...

🔍

☆

🔒 Incognito (2)

⋮

☰

Google Cloud

🔍

🔔

⋮

S

☑

Cloud Build / Triggers

Triggers

+ Create trigger

➔ Connect repository

🗑 Manage repositories

A Cloud Build trigger automatically starts a build whenever you make any changes to your source code or some other incoming event.

☰ Filter

Enter property name or value

?

☰

| Name ↑                                  | Region   | Description | Repository                                                                                                                       |       |
|-----------------------------------------|----------|-------------|----------------------------------------------------------------------------------------------------------------------------------|-------|
| <a href="#">hello-cloudbuild</a>        | us-west1 | –           |  <a href="#">BSSE23058/hello-cloudbuild-app</a> | Run ⋮ |
| <a href="#">hello-cloudbuild-deploy</a> | us-west1 | –           |  <a href="#">BSSE23058/hello-cloudbuild-env</a> | Run ⋮ |

'hello-cloudbuild-deploy' has been created on BSSE23058/hello-cloudbuild-env

×



8:31 PM

13-Oct-2025

## Dashboard

manually submitted builds, on the History page.

Filter Enter property name or value

Running: BSSE23058/hello-cloudbuild-app - hello-cloudbuild

| Latest Build                      | Durati... | Trigger description | Source | Commit                   |
|-----------------------------------|-----------|---------------------|--------|--------------------------|
| <a href="#">10/13/25, 8:34...</a> | 00:00:35  | —                   | ...    | <a href="#">1db9b...</a> |

Build History

[View all](#)

Average Duration ?

—

Pass - Fail % ?

—

Running: BSSE23058/hello-cloudbuild-env - hello-cloudbuild-deploy

| Latest Build                      | Durati... | Trigger description | Source | Commit                    |
|-----------------------------------|-----------|---------------------|--------|---------------------------|
| <a href="#">10/13/25, 8:34...</a> | 00:00:20  | —                   | ...    | <a href="#">010f63...</a> |

Build History

[View all](#)

Average Duration ?

—

Pass - Fail % ?

—

Build started on branch  
master for trigger hello-  
cloudbuild-deploy

Show



Cloud Build - qwiklabs-gcp-03

Google Cloud

qwiklabs-gcp-03-a83160aaeca9

Search (/) for resources, docs, products, and more

Cloud Build / Dashboard

Dashboard

manually submitted builds, on the History page.

Filter Enter property name or value

Successful: BSSE23058/hello-cloudbuild-app - hello-cloudbuild

Latest Build 10/13/25, 8:34 Durati... 00:00:38 Trigger description Source Commit 1db9b...

Build History View all

Average Duration 00:00:38

Pass - Fail % 100% - 0%

Failed: BSSE23058/hello-cloudbuild-env - hello-cloudbuild-deploy

Latest Build 10/13/25, 8:47 Durati... 00:00:22 Trigger description Source Commit 010f63... Failed step 1. Build

Build History View all

Average Duration 00:00:18

Pass - Fail % 0% - 100%

Recommended for you

- Build configuration overview
- Quickstart: Build
- Creating a basic build configuration file
- Creating and managing build triggers
- Cloud builders

22°C Clear

Search

Software Development & Con... Implement CI/CD Pipelines on GoogleCloudSkillsboost/Goog... Google Gemini

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Google Cloud

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Apply your skills in Google Cloud console Get started

DevSecOps > Course > Implement CI/CD Pipelines on Google Cloud >

10 Implement CI/CD Pipelines on Google Cloud

Course - 1 Complete hour 15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab 00:54:23

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username student-00-21b18e955885i

Password 36EUq9V9TSP

Project ID qwiklabs-gcp-03-a83160aaeca9

Previous

```
# unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

# [START cloudbuild]
steps:
# This step runs the unit tests on the app
- name: 'python:3.7-slim'
  id: Test
  entrypoint: /bin/sh
  args:
    - -c
    - 'pip install flask && python test_app.py -v'

# This step builds the container image.
- name: 'gcr.io/cloud-builders/docker'
  id: Build
  args:
    - 'build'
```

Lab instructions and tasks 85/100

GSP1077

Overview

Objectives

Setup and requirements

Task 1. Initialize your lab

Task 2. Create the Git repositories in GitHub repositories

Task 3. Create a container image with Cloud Build

Task 4. Create the Continuous Integration

Next

Software Development & Con...Implement CI/CD Pipelines on...GoogleCloudSkillsboost/Googl...Google Gemini

cloudskillsboost.google/paths/76/course\_templates/691/labs/594612#step8

Google CloudDashboardExplorePathsSubscriptions

Apply your skills in Google Cloud consoleGet started

10Implement CI/CD Pipelines on Google Cloud

Course - 1Completehour 15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

DevSecOps > Course > Implement CI/CD Pipelines on Google Cloud >

End Lab00:54:09

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Usernamestudent-00-21b18e955805i

Password36EUq9V9TSvp

Project IDqw1klabs-gcp-03-a83160ai

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# [START cloudbuild]

steps:

- name: 'python:3.7-slim'

id: Test

entrypoint: /bin/sh

args:

- -c

- 'pip install flask && python test\_app.py -v'

# This step builds the container image.

- name: 'gcr.io/cloud-builders/docker'

id: Build

args:

- 'build'

Build20 / 20

Create the Continuous Integration (CI) PipelineCheck my progress20 / 20

Accessing GitHub from a build via SSH keysCheck my progress20 / 20

Create the Test Environment and CD PipelineCheck my progress5 / 20

Task 2. Create the Git repositories in GitHub repositories

Task 3. Create a container image with Cloud Build

Task 4. Create the Continuous Integration

< Previous

Next >

22°C Clear

Search

8:47 PM13-Oct-2025